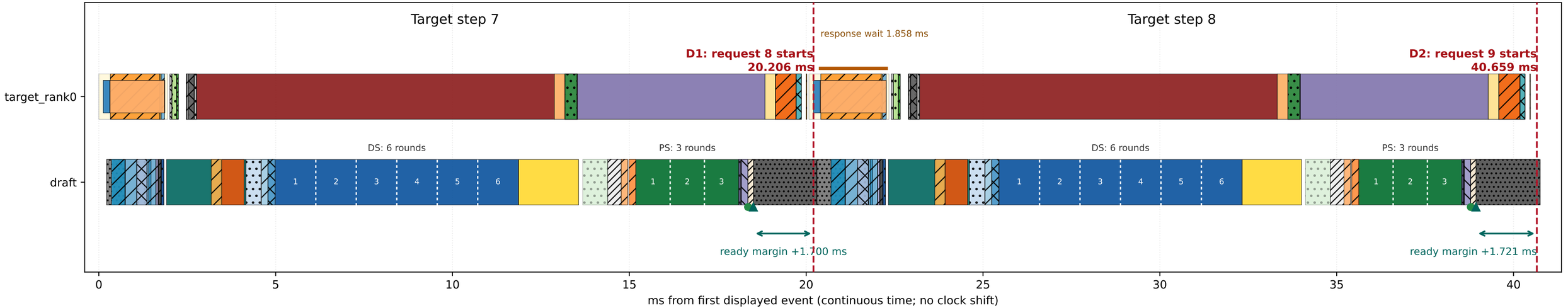
LayerSkip Llama2-7B + AMD-Llama-135M | Adopted PS optimization | P1 hit | UID 514_t0 | steps 7-8

Runtime matches ver1.2 9390e21 | RTX4090 | B1 BF16 T0.7 | exit20 K1/K2 6/3 M1/M2 12/8 C3 | frontier

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
7	18.356	18.506	20.206	+1.851	+1.700	0.189	1.858
8	38.792	38.938	40.659	+1.867	+1.721	0.184	1.536

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

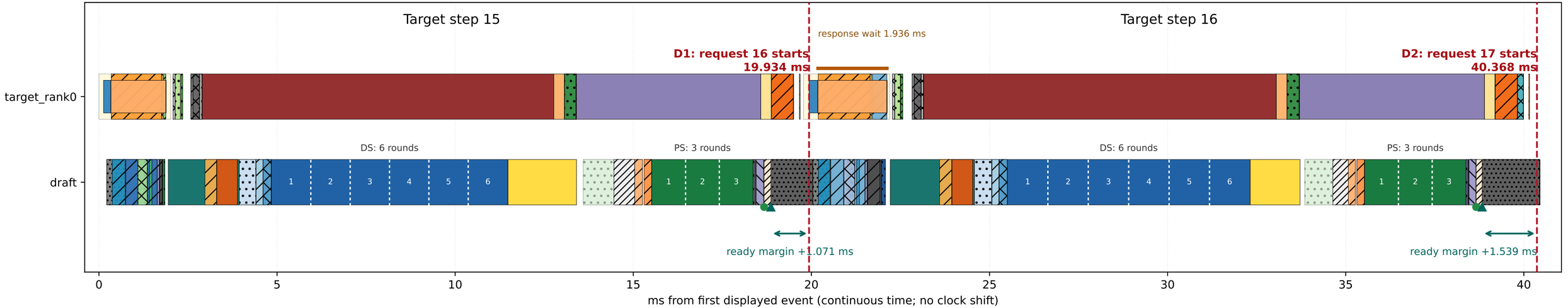
Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama2-7B + AMD-Llama-135M | Adopted PS optimization | P2 hit | UID 397_t0 | steps 15-16

Runtime matches ver1.2 9390e21 | RTX4090 | B1 BF16 T0.7 | exit20 K1/K2 6/3 M1/M2 12/8 C3 | frontier

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

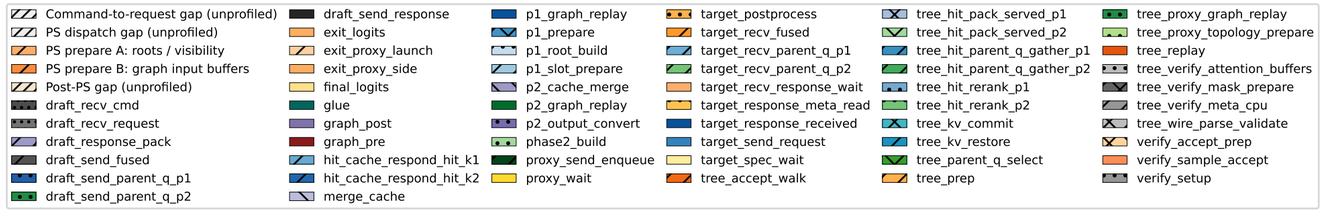
Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
15	18.672	18.863	19.934	+1.262	+1.071	0.241	1.936
16	38.661	38.829	40.368	+1.707	+1.539	0.204	1.589

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

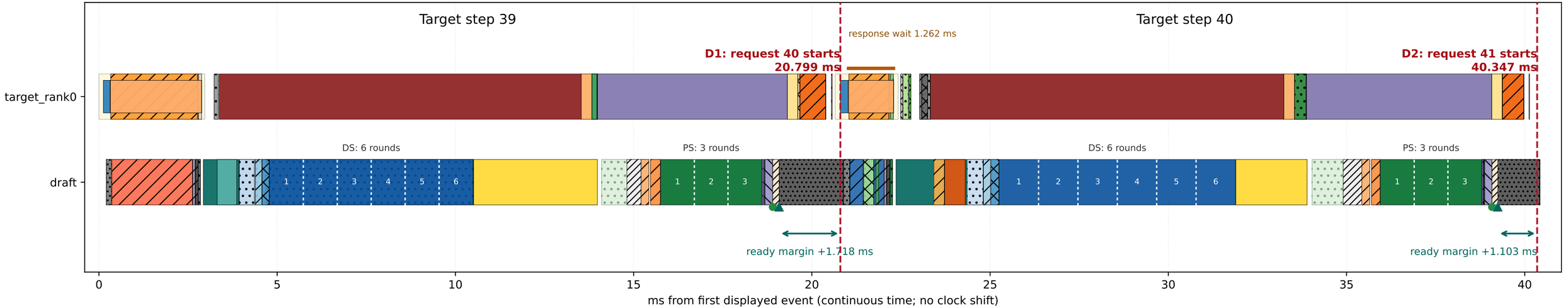
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama2-7B + AMD-Llama-135M | Adopted PS optimization | Miss | UID 234_t0 | steps 39-40

Runtime matches ver1.2 9390e21 | RTX4090 | B1 BF16 T0.7 | exit20 K1/K2 6/3 M1/M2 12/8 C3 | frontier

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
39	18.904	19.081	20.799	+1.895	+1.718	0.218	1.262
40	39.083	39.245	40.347	+1.265	+1.103	0.206	2.628

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.